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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT	FinTrack: Personal Expense Management System	
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Abstract

Managing personal finances is a significant challenge for individuals due to irregular spending habits, manual record keeping, limited financial awareness, and the absence of a centralized platform for tracking daily expenses. Users often find it difficult to maintain accurate records of their income and expenditures, categorize transactions, monitor budgets, and understand their spending patterns. Traditional methods such as notebooks and spreadsheets require considerable manual effort and may not provide timely insights into financial behavior. To address these challenges, this project proposes FinTrack: Personal Expense Management System, a web-based application designed to simplify and automate personal financial management.

The proposed system provides users with a centralized platform to record, categorize, update, and monitor their income and expenses. Users can enter transactions with details such as amount, date, category, payment method, and description. The system automatically calculates total income, total expenses, and available balance, providing users with a clear overview of their financial status. Budget management features enable users to set spending limits for different categories and monitor their progress toward those limits. The system can also provide alerts when expenses approach or exceed predefined budgets, helping users control unnecessary spending.

The portal provides an intuitive dashboard that presents financial information through interactive charts, graphs, summaries, and category-wise expense reports. These visualizations allow users to identify their major spending categories and analyze their financial habits over different time periods. Users can search and filter transactions based on dates, categories, and payment methods, making it easier to review and manage financial records. The system also

supports secure user authentication and maintains individual financial data separately to ensure privacy and controlled access.

The proposed application can be developed using React.js for the frontend, Spring Boot (Java) for the backend, and MySQL for database management. RESTful APIs facilitate communication between the frontend and backend, while secure authentication mechanisms protect user accounts and financial information. Development practices such as Git and GitHub can be incorporated for version control, while validation and secure coding practices help maintain system reliability. The modular architecture allows future integration of advanced features such as automated expense categorization, financial recommendations, and predictive spending analysis.

The portal primarily supports three major stakeholders: Users, Financial Administrators, and System Administrators. Users can manage their personal transactions, budgets, and financial reports. Financial administrators can monitor system-level financial management features and generate aggregated reports where applicable, while system administrators can manage users, maintain application configurations, and ensure system security and availability. The proposed system aims to reduce manual financial record keeping, improve budgeting discipline, increase financial awareness, and support informed decision-making through centralized and visualized expense management.

The proposed system is evaluated based on usability, accuracy, security, responsiveness, data consistency, and reliability. By providing an organized platform for recording and analyzing personal finances, FinTrack demonstrates the practical application of Software Engineering, Full-Stack Development, Database Management, Web Technologies, and Data Visualization concepts in personal financial management.